

★ Rating changes for last rounds are temporarily rolled back. They will be returned soon.

A. Shortest Path

time limit per test: 1 second

memory limit per test: 1024 megabytes

You are given an **directed weighted** graph with N nodes and M edges. The nodes are numbered from 1 to N . The graph contains no self-loops or multiple edges.

There is a source and a destination. Your task is to find the shortest distance from the source node to the destination node and print the path taken. If multiple shortest paths exist, print any one of them. If no such path exists, print -1 .

Input

The first line contains four integers

N, M, S, D ($2 \leq N \leq 2 \times 10^5, 1 \leq M \leq 3 \times 10^5, 1 \leq S, D \leq N$) — the number of vertices, total number of edges, source and destination.

The second line contains M integers $u_1, u_2, u_3 \dots u_m$ ($1 \leq u_i \leq N$) — where the i -th integer represents the node that is one endpoint of the i -th edge.

The third line contains M integers $v_1, v_2, v_3 \dots v_m$ ($1 \leq v_i \leq N$) — where the i -th integer represents the node that is the other endpoint of the i -th edge.

The fourth line contains M integers $w_1, w_2, w_3 \dots w_m$ ($1 \leq w_i \leq 10^6$) — where the i -th integer represents the weight of the i -th edge.

The i -th edge of this graph is from the i -th node in the second line to the i -th node in the third line.

Output

- If a valid path exists from S to D , print the shortest distance S to D on the first line.
- On the second line, print the nodes in the path in order from S to D . If multiple shortest paths exist, print any one of them.
- If no such path exists, print -1 .

Examples

input

Copy

```
4 3 4 2
1 3 4
2 2 3
3 4 5
```

output

Copy

```
9
4 3 2
```

input

Copy

```
6 5 1 5
1 4 1 6 4
2 1 6 2 6
3 3 4 3 4
```

CSE221 Assignment 07 Summer 2026

Contest is running

5 days

Contestant



→ Submit?

Language: Java 21 64bit ▼

Choose file: Choose File No file chosen



Submit

output

Copy

-1**input**

Copy

2 1 2 1
2
1
7**output**

Copy

7
2 1**input**

Copy

5 7 2 4
1 1 5 4 2 3 2
5 4 3 5 5 4 3
3 8 2 6 6 4 3**output**

Copy

7
2 3 4

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